

RYAN LIU

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Education

Princeton University

PhD in Computer Science

Princeton, NJ

Aug 2023 – Present

Carnegie Mellon University

Fifth Year MS in Computer Science

Pittsburgh, PA

Aug 2022 – May 2023

BS in Computer Science, College and University Honors

Aug 2018 – May 2022

Research Experience

Princeton University (Advisor: [Thomas L. Griffiths](#))

Aug 2023 – Present

I study how large language models align with the patterns underlying human behavior. I also investigate applications surrounding the various representations of people that these models afford.

Carnegie Mellon University (Advisor: [Nihar B. Shah](#))

Dec 2019 – May 2022

I studied how to improve scientific peer review, addressing both reviewer biases (anchoring, citation bias) and improving reviewer-paper match quality. Results have been employed or informed the design at top conferences such as NeurIPS.

Publications

1. Addison J. Wu*, **Ryan Liu***, Shuyue Stella Li, Yulia Tsvetkov, and Thomas L. Griffiths. “**Ads in AI Chatbots? An Analysis of How Large Language Models Navigate Conflicts of Interest**” Preprint, 2026.
2. Addison J. Wu*, **Ryan Liu***, Xuechunzi Bai, and Thomas L. Griffiths. “**Large Language Models Develop Novel Social Biases Through Adaptive Exploration**” ICML 2026, Oral.
3. **Ryan Liu***, Dilip Arumugam, Cedegao E Zhang, Sean Escola, Xaq Pitkow, and Thomas L. Griffiths. “**Cognitive Models and AI Algorithms Provide Templates for Designing Language Agents**” Preprint, 2026.
4. Addison J. Wu*, **Ryan Liu***, Kerem Oktar*, Theodore R. Sumers, and Thomas L. Griffiths. “**Are Large Language Models Sensitive to the Motives Behind Communication?**” NeurIPS 2025.
5. Jiayi Geng*, Howard Chen*, **Ryan Liu***, Manoel Horta Ribeiro, Robb Willer, Graham Neubig, and Thomas L. Griffiths. “**Accumulating Context Changes the Beliefs of Language Models**” Preprint, 2025.
6. Katherine M. Collins, Cedegao E. Zhang, Graham Todd, Lance Ying, Mauricio Barba da Costa, **Ryan Liu**, Prafull Sharma, Adrian Weller, Ionatan Kuperwajs, Lionel Wong, Joshua B. Tenenbaum, and Thomas L. Griffiths. “**Evaluating Language Models’ Evaluations of Games**” Preprint, 2025.
7. **Ryan Liu***, Jiayi Geng*, Addison J. Wu, Ilia Sucholutsky, Tania Lombrozo, and Thomas L. Griffiths. “**Mind Your Step (by Step): Chain-of-Thought can Reduce Performance on Tasks where Thinking Makes Humans Worse**” ICML 2025.
8. Jacy Reese Anthis, **Ryan Liu**, Sean M Richardson, Austin C Kozlowski, Bernard Koch, James Evans, Erik Brynjolfsson, Michael Bernstein. “**LLM Social Simulations Are a Promising Research Method**” ICML Position Paper Track 2025.
9. **Ryan Liu***, Jiayi Geng*, Joshua C. Peterson, Ilia Sucholutsky, and Thomas L. Griffiths. “**Large Language Models Assume People are More Rational than We Really are**” ICLR 2025.
10. **Ryan Liu**, Howard Yen, Raja Marjieh, Thomas L. Griffiths, and Ranjay Krishna. “**Improving Interpersonal Communication by Simulating Audiences with Language Models**” CogSci 2025.
11. Kaiqu Liang, Haimin Hu, Ryan Liu, Thomas L Griffiths, Jaime Fernández Fisac. “**RLHS: Mitigating Misalignment in RLHF with Hindsight Simulation**” Preprint, 2025.
12. Alexander Ku, Declan Campbell, Xuechunzi Bai, Jiayi Geng, Ryan Liu, Raja Marjieh, R Thomas McCoy, Andrew Nam, Ilia Sucholutsky, Veniamin Veselovsky, Liyi Zhang, Jian-Qiao Zhu, Thomas L Griffiths. “**Using the Tools of Cognitive Science to Understand Large Language Models at Different Levels of Analysis**” Preprint, 2025.

13. Lance Ying, Katherine M Collins, Lionel Wong, Ilia Sucholutsky, Ryan Liu, Adrian Weller, Tianmin Shu, Thomas L Griffiths, Joshua B Tenenbaum. **“On Benchmarking Human-like Intelligence in Machines”** Preprint, 2025.
14. Tongshuang Wu, Haiyi Zhu, Maya Albayrak, Alexis Axon, Amanda Bertsch, Wenxing Deng, Ziqi Ding, Bill Guo, Sireesh Gururaja, Tzu-Sheng Kuo, Jenny T Liang, **Ryan Liu**, Ihita Mandal, Jeremiah Milbauer, Xiaolin Ni, Namrata Padmanabhan, Subhashini Ramkumar, Alexis Sudjianto, Jordan Taylor, Ying-Jui Tseng, Patricia Vaidos, Zhijin Wu, Wei Wu, Chenyang Yang. **“LLMs as Workers in Human-Computational Algorithms? Replicating Crowdsourcing Pipelines with LLMs”** CHI Conference on Human Factors in Computing Systems, Extended Abstract, 2025.
15. **Ryan Liu***, Theodore R. Sumers*, Ishita Dasgupta, and Thomas L. Griffiths. **“How do Large Language Models Navigate Conflicts between Honesty and Helpfulness?”** Oral, ICML 2024.
16. **Ryan Liu** and Nihar B. Shah. **“ReviewerGPT? An Exploratory Study on Using Large Language Models for Paper Reviewing”** Oral, AAAI SDU 2024.
17. **Ryan Liu**, Steven Jecmen, Fei Fang, Vincent Conitzer, and Nihar B. Shah. **“Testing for Reviewer Anchoring in Peer Review: A Randomized Controlled Trial”** PLOS One, 2024.
18. Yihan Cao*, Shuyi Chen*, **Ryan Liu***, Zhiruo Wang, and Daniel Fried. **“API-Assisted Code Generation for Question Answering on Varied Table Structures”** EMNLP 2023.
19. Ivan Stelmakh, Charvi Rastogi, **Ryan Liu**, Shuchi Chawla, Federico Echenique, and Nihar B. Shah. **“Cite-seeing and Reviewing: A Study on Citation Bias in Peer Review”** PLOS One, 2023.
20. Steven Jecmen, Hanrui Zhang, **Ryan Liu**, Fei Fang, Vincent Conitzer, and Nihar B. Shah. **“Near-Optimal Reviewer Splitting in Two-Phase Paper Reviewing and Conference Experiment Design”** AAAI HCOMP 2022, Best Paper Honorable Mention.
21. Steven Jecmen, Hanrui Zhang, **Ryan Liu**, Nihar B. Shah, Vincent Conitzer, and Fei Fang. **“Mitigating Manipulation in Peer Review via Randomized Reviewer Assignments”** NeurIPS 2020.

Work Experience

Assistant in Instruction, Princeton COS 598B Machine Behavior Jan – May 2025

I mentored and held office hours for ten course projects exploring the field of human-AI interaction and LLM behavior.

Assistant in Instruction, Princeton COS 350 Ethics in Computing Sep – Dec 2024

I taught precepts, developed homework material, graded assignments, and held office hours to support student learning.

ACM Conference on Economics and Computation Workflow Chair Nov 2022 – Mar 2023

I developed and implemented the algorithm for paper-reviewer matching at EC’23 and conducted various analyses.

Facebook AI/ML Software Engineering Intern May – August 2021

I built an LSTM using Pytorch on 200 GB irregular time series data to predict future eye gaze positions for the Oculus headset.

Teaching Assistant, CMU 15-112 Fundamentals of Programming Jul – Dec 2019

I mentored 10 students’ individual projects, with 2 mentees scoring in the top 6 of 400+ students, and created and taught a quantum computing special interest lecture, with Rigetti sponsoring \$5000 compute for a live demo.

Invited Talks/Presentations

UCLA NLP Seminar Dec 2025

The Psychology of AI Agents

Stanford Politics and Social Change Lab Apr 2025

Predicting and Simulating New People using Existing Agents

Stanford NLP Seminar Jan 2025

Mind Your Step (by Step): Chain-of-Thought can Reduce Performance on Tasks where Thinking Makes Humans Worse

Stanford HCI Group Jan 2025

Predicting and Simulating New People using Existing Agents

University of Texas at Austin NLP Group Nov 2024

Mind Your Step (by Step): Chain-of-Thought can Reduce Performance on Tasks where Thinking Makes Humans Worse

Oral, International Conference on Machine Learning 2024 Jul 2024

How do Large Language Models Navigate Conflicts between Honesty and Helpfulness?

Oral, AAAI SDU Workshop 2024 ReviewerGPT? An Exploratory Study on Using Large Language Models for Paper Reviewing	Feb 2024
London Machine Learning Meetup (Role: Moderator) Joon Sung Park — Generative Agents: Interactive Simulacra of Human Behavior	Aug 2023
Data Skeptics Podcast Automated Peer Review	Jul 2023
Carnegie Mellon University Meeting of the Minds Identifying Human Biases in Peer Review via Real-Subject Experiments	May 2022
Carnegie Mellon University Meeting of the Minds Improving Algorithmic Tools for Conference Peer Review Research	May 2021
Carnegie Mellon University Fall Undergraduate Research Showcase Creating Robustness within Conference Peer Review	Dec 2020
Carnegie Mellon University Meeting of the Minds Assignment Algorithms to Prevent Quid-Pro-Quo in Conference Peer Review	May 2020

Mentoring

Jiayi Geng (Princeton) → PhD student at CMU	ICLR 2025, ICML 2025
Sunny Chunhua Yu (Stanford) → PhD student at Princeton	Current
Addison J. Wu	ICML 2026 Oral, ICML 2025, COLM 2025 (Workshop), NeurIPS 2025, NeurIPS 2025 (Workshop) Current

Academic Honors & Professional Services

Reviewer, International Conference on Learning Representations 2026
 Reviewer, Conference on Neural Information Processing Systems 2025
 Reviewer, International Conference on Learning Representations 2025
 Reviewer, International Conference on Learning Representations Bidirectional Alignment Workshop 2025
 Reviewer, Conference on Human Factors in Computing Systems 2025
 Reviewer, Conference on Neural Information Processing Systems Behavioral ML Workshop 2024
 Reviewer, Conference on Neural Information Processing Systems 2024
 Student Organizer, Decentralized Social Media Workshop 2024
 Workflow Chair, ACM Conference on Economics and Computation 2023
 National Science Foundation Research Experience for Undergraduates Grant 2021
 Carnegie Mellon University Summer Undergraduate Research Apprenticeship 2020